

100% stacked Bar Chart

The School Pizza Party Story

Imagine there are 3 classes:

- Class 9A
- Class 9B
- Class 9C

After a pizza party, we ask students:

"Did you like the pizza?"

Possible answers:

- 😊 Happy
 - 😐 Okay
 - 😞 Didn't Like It
-

Situation 1: Normal Bar Chart

Let's say:

Class 9A

50 students

- Happy = 40
 - Okay = 5
 - Sad = 5
-

Class 9B

20 students

- Happy = 10
 - Okay = 5
 - Sad = 5
-

Class 9C

10 students

- Happy = 5
 - Okay = 2
 - Sad = 3
-

Now ask students:

Which class has the most happy students?

Answer:

9A

Because:

$40 > 10 > 5$

Simple.

The Problem

Ask:

Is 9A really happier?

Or

Does 9A just have more students?

Students start thinking.

Because:

9A has 50 students.

9C has only 10 students.

Comparing totals is unfair.

It's like comparing:



Elephant

and



Mouse

just by weight.

Of course the elephant wins.

Enter The Superhero

100% Stacked Bar Chart

Now tell students:

"Let's forget how many students are in each class."

Instead ask:

"Out of every 100 students, how many were happy?"

Now everything becomes fair.

Convert To Percentages

Class 9A

40 Happy out of 50

= 80%

Class 9B

10 Happy out of 20

= 50%

Class 9C

5 Happy out of 10

= 50%

Now every class gets:

One Equal-Sized Bar

Because every bar means:

100% of that class

Imagine three identical chocolate bars.

Each chocolate bar represents one class.

Class 9A



😊 Happy = 80%

☹ Sad = 20%

Class 9B



😊 Happy = 50%

☹ Sad = 50%

Class 9C



😊 Happy = 50%

☹ Sad = 50%

Now we can compare fairly.

The Magic Question

Ask students:

"If all classes were the same size, which class would be happiest?"

Now answer:

9A

Because:

80% happy.

Why Use 100% Stacked Charts?

Normal Chart answers:

"How many?"

Example:

How many students liked pizza?

100% Stacked Chart answers:

"What percentage?"

Example:

What fraction of students liked pizza?

Real-Life Example

Ask:

Which school performed better?

School A

1000 students

900 passed

School B

100 students

95 passed

Normal chart:

900 > 95

School A looks better.

Percentage chart:

School A = 90%

School B = 95%

Now School B looks better.

One-Line Rule

Write this on the board:

Normal Bar Chart = Compare Quantities

100% Stacked Bar Chart = Compare Percentages

That's the easiest way for Class 9 students to understand it.

Classroom Activity

Ask students:

Favourite Sport

- Cricket
- Football
- Badminton

Collect data from:

- Class 9A
- Class 9B
- Class 9C

First draw a normal bar chart.

Then convert to percentages and draw a 100% stacked bar chart.

Students will immediately see why percentages are sometimes more useful than totals.

This hands-on activity usually makes the concept click within 5 minutes.

